A particularly nasty problem for computer speed is *backward compatibility*. When a computer company introduces a new product, say a data acquisition card or a software program, they want to sell it into the largest possible market. This means that it must be compatible with most of the computers currently in use, which could span several generations of technology. This frequently limits the performance of the hardware or software to that of a much older system. For example, suppose you buy an I/O card that plugs into the bus of your 200 MHz Pentium personal computer, providing you with eight digital lines that can transmit and receive data one byte at a time. You then write an assembly program to rapidly transfer data between your computer and some external device, such as a scientific experiment or another computer. Much to your surprise, the maximum data transfer rate is only about 100,000 bytes per second, more than *one thousand* times slower than the microprocessor clock rate! The villain is the ISA bus, a technology that is *backward compatible* to the computers of the early 1980s.

Table 4-6 provides execution times for several generations of computers. Obviously, you should treat these as very rough approximations. If you want to understand *your* system, take measurements on *your* system. It's quite easy; write a loop that executes a *million* of some operation, and use your watch to time how long it takes. The first three systems, the 80286, 80486, and Pentium, are the standard desk-top personal computers of 1986, 1993 and 1996, respectively. The fourth is a 1994 microprocessor designed especially for DSP tasks, the Texas Instruments TMS320C40.

	80286 (12 MHz)	80486 (33 MHz)	PENTIUM (100 MHz)	TMS320C40 (40 MHz)
INTEGER				
A% = B%+C%	1.6	0.12	0.04	
A% = B%–C%	1.6	0.12	0.04	
A% = B%×C%	2.7	0.59	0.13	
A% = B%÷C%	64	9.2	1.5	
FLOATING POINT				
A = B+C	33	2.5	0.50	0.10
A = B–C	35	2.5	0.50	0.10
A = B×C	35	2.5	0.50	0.10
A = B÷C	49	4.5	0.87	0.80
A = SQR(B)	45	5.3	1.3	0.90
A = LOG(B)	186	19	3.4	1.7
A = EXP(B)	246	25	5.5	1.7
A = B^C	311	31	5.3	2.4
A = SIN(B)	262	30	6.6	1.1
A = ARCTAN(B)	168	21	4.4	2.2

TABLE 4-6
Measured execution times for various computers. Times are in microseconds. The 80286, 80486, and Pentium are three generations of personal computers, while the TMS320C40 is a microprocessor specifically designed for DSP tasks. All of the personal computers include a math coprocessor. Use these times only as a general estimate; times on your computer will vary according to the particular hardware and software used.